1. By converting the daily return into a weekly return for each firm, we obtain:

Firm	**Expected Return**	**Standard Deviation**
Canadian National Railway Company(CNR)	0.567%	3.99%
Fortis Inc.(FTS)	0.178%	2.14%
RBC(RY)	0.311%	2.00%
Suncor Energy(SU)	0.259%	3.36%
Telus(T)	0.248%	2.01%
Thomson Reuters Company(TRI)	0.389%	2.07%

Table 1: Equity vs. Expected Weekly Return and Std

and the covariance matrix

	CNR	FTS	RY	SU	T	TRI
CNR	0.159%	0.015%	0.016%	0.026%	0.010%	0.019%
FTS	0.015%	0.046%	0.014%	0.018%	0.015%	0.011%
RY	0.016%	0.014%	0.040%	0.031%	0.011%	0.016%
SU	0.026%	0.018%	0.031%	0.113%	0.012%	0.023%
T	0.010%	0.015%	0.011%	0.012%	0.040%	0.012%
TRI	0.019%	0.011%	0.016%	0.023%	0.012%	0.043%

Table 2: Covariance Matrix of Weekly Returns of Each Stock

In the subsequent questions, we will use the tickers to represent each stock/equity.

2. To find the optimal portfolio, we first converted the yearly return rate of the 3-month risk free T-Bill into a 3-month rate using the formula

$$r_m = \frac{90}{365} r_y = \frac{90}{365} \cdot 0.01 \approx 0.0024658$$

Then, assuming compounding and there are 12 weeks in 3 months, we obtain the weekly return as

$$r_w = (1 + r_m)^{\frac{1}{12}} \approx (1 + 0.0024658)^{\frac{1}{12}} \approx 0.0.020525\%$$

where r_y, r_m, r_w are the annual, 3 months, and the weekly return, respectively. Then using this risk-free rate, we aim to maximize the function

$$S(w_1, ..., w_6) = \frac{\sum_{i=1}^{6} w_i E[r_i] - 0.0.020525\%}{\left[\sum_{i=1}^{6} w_i^2 \sigma_i^2\right] + \left[\sum_{i \neq j} w_i w_j Cov(r_i, r_j)\right]}$$

where $E[r_i]$ and σ is the expected return and the standard deviation of the ith stock in Table 1, and $Cov(r_i, r_j)$ is the covariance of the weekly return between stock i and stock j as shown in Table. The constraint of the optimization is

$$\sum_{i=1}^{6} w_i = 1.$$

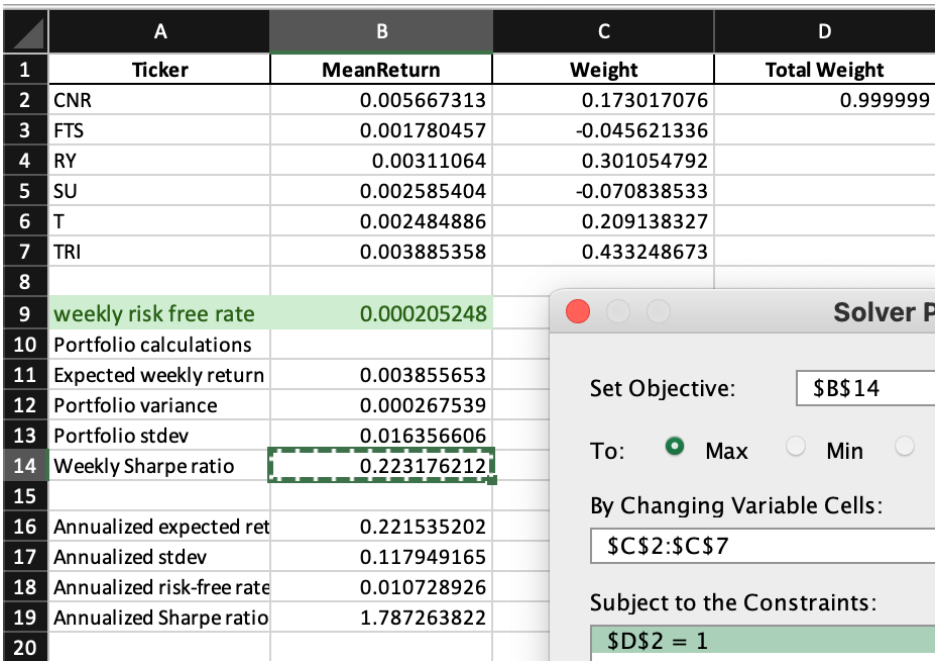
But by inspection we see that the denominator of S(*i.e.* the variance of the risky portfolio) can be expressed as a product of matrix and vectors:

$$\mathbf{w}^T \sum \mathbf{w}$$

where $\mathbf{w}$ is the column vector representing the weights associated with each equity, and $\sum$ is the covariance matrix in Table 2.
Hence, we equivalently want to maximize

$$\frac{\sum_{i=1}^{6} w_1 \mathbb{E}[r_i] - 0.0.020525\%}{\mathbf{w}^T \sum \mathbf{w}}$$

subject to $\sum_{i=1}^{6} w_i = 1$.
Practically, we solve this using the solver function provided in Excel:



	A	B	C	D
1	Ticker	MeanReturn	Weight	Total Weight
2	CNR	0.005667313	0.173017076	0.999999
3	FTS	0.001780457	-0.045621336	
4	RY	0.00311064	0.301054792	
5	SU	0.002585404	-0.070838533	
6	T	0.002484886	0.209138327	
7	TRI	0.003885358	0.433248673	
8				
9	weekly risk free rate	0.000205248		
10	Portfolio calculations			
11	Expected weekly return	0.003855653		
12	Portfolio variance	0.000267539		
13	Portfolio stdev	0.016356606		
14	Weekly Sharpe ratio	0.223176212		
15				
16	Annualized expected ret	0.221535202		
17	Annualized stdev	0.117949165		
18	Annualized risk-free rate	0.010728926		
19	Annualized Sharpe ratio	1.787263822		
20				

Figure 1: Demonstration of Numerically Optimizing Sharpe Ratio In Excel

In this spreadsheet,

(a) weekly risk-free return(B9) is highlighted in green, and is a constant that we calculated earlier using the return rate on the 3-Month T-Bill,
(b) expected weekly return(B11) is defined using the formula SUMPRODUCT(B2:B7, C2:C7),
(c) portfolio variance(B12) is defined using the formula MMULT(TRANSPOSE(C2:C7), MMULT(CoVar!B2:G7, C2:C7)) where CoVar!B2:G7 is the covariance matrix as discussed above,
(d) weekly sharpe ratio(B14) is defined using the formula (B11-B9)/B13,
(e) total weight(D2) is defined using the formula SUM(C2:C7)

Using the solver, we then maximize B14(*i.e.* the sharpe ratio) with variables C2:C7(the weights) subject to the constraint D2 = 1(sum of weights equals one), which then automatically updates every pre-defined cell in this picture once we obtain an optimal solution, as shown below:

	CNR	FTS	RY	SU	T	TRI
Weights	0.17	-0.05	0.30	-0.07	0.21	0.43

Table 3: Short Selling Allowed Portfolio Weights

and the corresponding statistics of the portfolio:

	Expected Return	Variance	Std Dev	Sharpe Ratio
Value	0.38557%	0.02675%	1.63567%	0.22318

Table 4: Short Selling Allowed Portfolio Weekly Statistics

3. Here is the $E(r) - \sigma$ plot of the six equities and the optimal portfolio constructed above using the six equities:

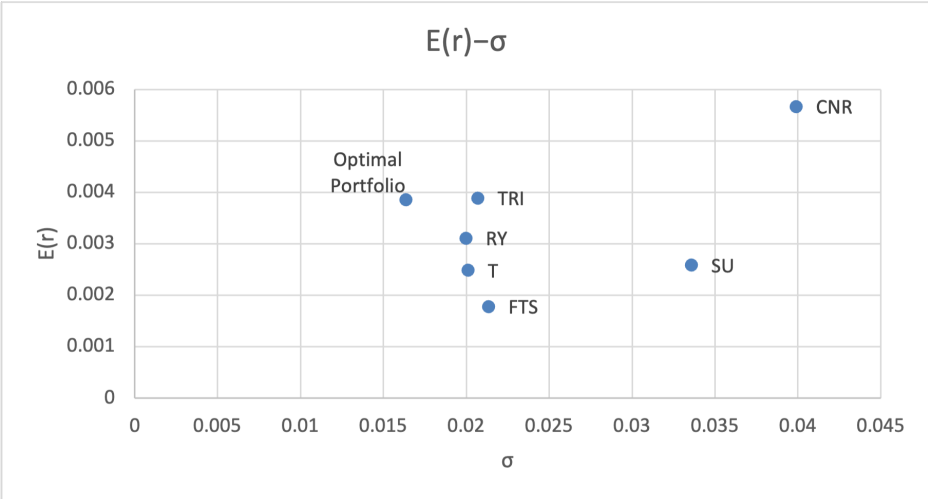


Figure 2: $E(r) - \sigma$ Plot of the Equities and the Optimal Risky Portfolio

Yes, the location of the optimal risky portfolio makes sense relative to the six equities because diversification lets us combine these risky stocks in a way that reduces overall volatility while keeping return high. In the plot, the optimal portfolio lies to the left of all the individual equities on the σ-axis, meaning it has lower risk than holding any single stock alone, and it lies above most of them on the $E(r)$ axis, meaning it offers a higher expected return than most individual equities. No single equity has both a higher return and lower risk than this portfolio (for example, TRI and RY have similar returns but higher σ), which is exactly what we expect from the optimal risky portfolio on the efficient frontier.

4. i) Lower covariance means greater diversification, which lowers overall portfolio risk because assets that move less together reduce variance more effectively when combined. As a result, the efficient frontier shifts upward and to the left, offering higher returns for the same level of risk or lower risk for the same level of return. This also raises the Sharpe ratio of the optimal risky portfolio, since the tangency portfolio becomes steeper when diversification improves. In addition, the optimizer no longer needs to concentrate weights heavily in a few low-covariance assets, allowing weights

to be distributed more evenly across the six equities. Overall, lower covariances make the optimal risky portfolio safer, more diversified, and higher in Sharpe ratio.

ii) If the interest rate on T-bills were higher, it would change the capital allocation line but not the composition of the optimal risky portfolio itself. The weights of the optimal risky portfolio depend only on expected excess returns relative to one another and on the covariance matrix, so although a higher risk-free rate raises all excess returns' baseline, it does not change the relative structure that determines the tangency portfolio. What does change is the Sharpe ratio: as the risk-free rate increases, the term $E(r_p) - r_f$ becomes smaller, causing the Sharpe ratio of the optimal risky portfolio to fall. With a flatter CAL, investors allocate less wealth to the risky portfolio and more to T-bills, because the reward for taking on additional risk becomes relatively less attractive. Thus, a higher T-bill rate leaves the composition of the optimal risky portfolio unchanged but lowers its Sharpe ratio and reduces investors' demand for risky assets.

5. Here are the weights of each equity in the Minimum Variance Portfolio:

	CNR	FTS	RY	SU	T	TRI
Weights	0.04	0.20	0.26	-0.01	0.28	0.23

Table 5: Global Minimum Variance (GMV) Portfolio Weights

and the corresponding statistics of the portfolio:

	Expected Return	Variance	Std Dev	Sharpe Ratio
Value	0.29469%	0.020094%	1.41752%	0.19341

Table 6: GMV Portfolio Statistics

Including the GMV portfolio in the $E(r) - \sigma$ plot, we obtain

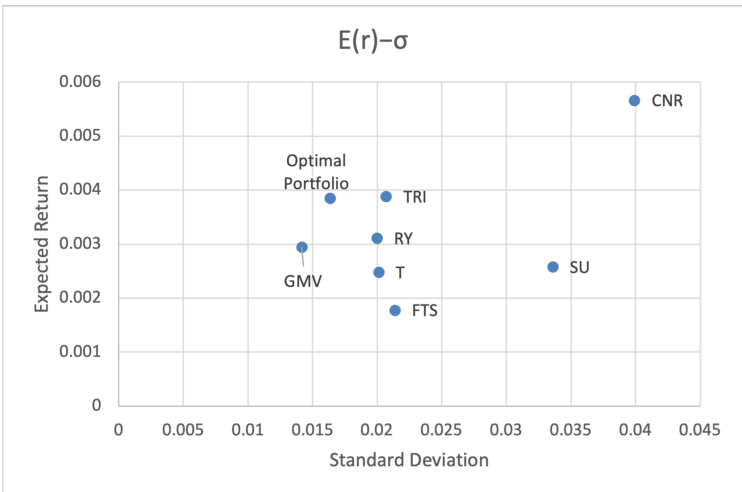


Figure 3: $E(r) - \sigma$ Plot of the Stocks, the Optimal Risky and the GMV Portfolio

6. To do so, we simply add the constraint that all variables must be non-negative in the solver. Then, imitating what we did in question 2, we obtain

	CNR	FTS	RY	SU	T	TRI
Weights	0.16	0.00	0.24	0.00	0.19	0.40

Table 7: No Short Selling Portfolio Weights

and the corresponding statistics of the portfolio follows

	Expected Return	Variance	Std Dev	Sharpe Ratio
Value	0.37240%	0.02536%	1.59259%	0.22094

Table 8: No Short Selling Portfolio Statistics

Incorporating this portfolio into our plot:

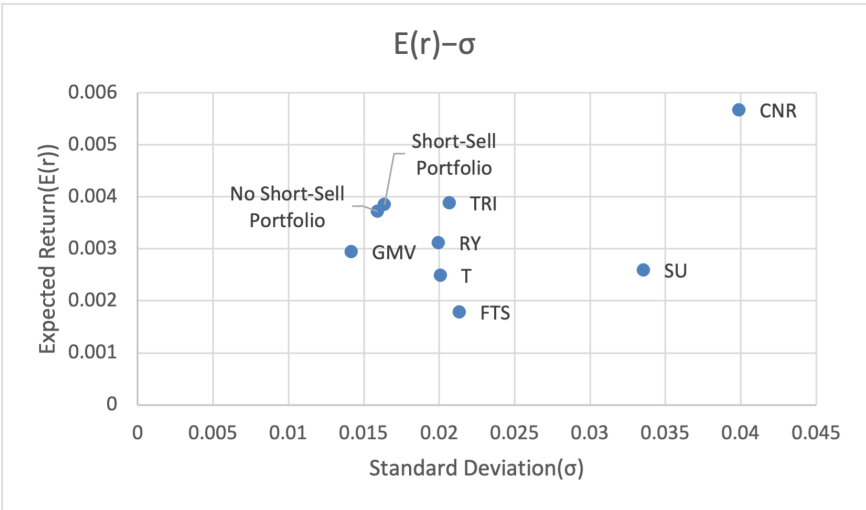


Figure 4: $E(r) - \sigma$ Plot of the Stocks, the Short-Sell/No Short-Sell Risky and the GMV Portfolio

7. Here is a table comparing the short selling allowed portfolio and the no short selling portfolio:

Portfolio	**Expected Return**	**Variance**	**Std Dev**	**Sharpe Ratio**
Short Selling Allowed	0.38557%	0.02675%	1.63567%	0.22318
No Short Selling	0.37240%	0.02536%	1.59259%	0.22094
Change [1]	-3.416%	-5.196%	-2.634%	-1%

Table 9: Comparison of Portfolio Statistics: Short Selling Allowed vs. No Short Selling

The no-short-selling requirement did not have a large effect on the optimal risky portfolio. Although forbidding short sales does change the feasible set of portfolios, the

[1] Change measures the percentage change when moving from short selling allowed to no short selling.

resulting impact on the portfolio's performance in our case is small. The expected return falls only 3.416%, and the portfolio's standard deviation also decreases only marginally by 2.634%. Because both expected return and risk decline by similar small proportions, the Sharpe ratio hardly changes at all. It fell from 0.22318 to 0.22094, a 1% difference. This indicates that the short-selling constraint does not materially weaken the portfolio's risk–return tradeoff. Overall, the optimal risky portfolios with and without short selling are very similar, meaning the no-short-selling requirement had only a modest effect.

8. Let y be the weights of the risky portfolio, r_{nss} be the expected return on the no short-selling risky portfolio, r_f be the risk-free return, and σ_{nss} be the standard deviation of the no short-selling risky portfolio. Then using the formula

$$y = \frac{r_{nss} - r_f}{A\sigma_{nss}^2} = \frac{0.37240\% - 0.020525\%}{15 \cdot 0.02536\%} \approx 0.925 \Rightarrow (1-y) \approx 0.075.$$

Thus, about 92.5% of the wealth should go into the risky portfolio, while 7.5% of the wealth will be invested in the 3-month risk-free T-Bill. With a total investment amount of \$1,000,000, this corresponds to roughly \$925,000 invested in the risky portfolio and about \$75,000 invested in the T-bill.

The \$925,000 allocated to the risky portfolio is then distributed across the six equities according to their no-short-selling tangency weights, resulting in approximately \$148,000 in CNR, \$222,000 in RY, \$175,750 in T, and \$370,000 in TRI with zero allocations to FTS and SU.

To find the expected return and the variance on the optimal portfolio, we use

$$E(r) = yE(r_{nss}) + (1-y)r_f = 0.925(0.37240\%) + 0.075(0.020525\%) \approx 0.34601\%$$

and

$$\sigma^2 = y^2\sigma_{nss}^2 = 0.925^2 \cdot 0.02536\% \approx 0.0217\%$$

so therefore $\sigma = \sqrt{0.0217\%} \approx 1.47309\%$.
Substituting, it follows that the utility acquired from this portfolio is

$$U = 0.34601\% - \frac{15 \cdot 0.0217\%}{2} \approx 0.0018326.$$

9. Let y be the weights of the short-selling risky portfolio, r_{ss} be the expected return on the short-selling risky portfolio, r_f be the risk-free return, and σ_{ss} be the standard deviation of the short-selling risky portfolio. Then using the formula

$$y = \frac{r_{ss} - r_f}{A\sigma_{ss}^2} = \frac{0.38557\% - 0.020525\%}{15 \cdot 0.02675\%} \approx 0.90977 \Rightarrow (1-y) \approx 0.09023.$$

Thus, about 90.977% of the wealth should go into the risky portfolio, while 9.023% of the wealth will be invested in the 3-month risk-free T-Bill. With a total investment amount of \$1,000,000, this corresponds to roughly \$909,770 invested in the short sellable portfolio and about \$90,230 invested in the T-bill.

The \$909,770 allocated to the short sellable portfolio is then distributed across the six equities according to their tangency weights, resulting in approximately \$154,660.9 in CNR, \$272,931 in RY, \$191,051.7 in T, \$391,201.1 in TRI, while borrowing \$45,488.5 from FTS, and \$63,683.9 from SU.

To find the expected return and the variance on this very short-selling-allowed optimal portfolio, we use

$$E(r) = yE(r_{ss}) + (1-y)r_f = 0.90977(0.38557\%) + 0.09023(0.020525\%) \approx 0.35263\%$$

and

$$\sigma^2 = y^2\sigma_p^2 = 0.90977^2 \cdot 0.02675\% \approx 0.02214\%$$

so therefore $\sigma = \sqrt{0.02214\%} \approx 1.48795\%$.

10. Substituting the expected return and the variance calculated from the previous question, it follows that the utility acquired from the short sellable optimal portfolio is

$$U = 0.35263\% - \frac{15 \cdot 0.02214\%}{2} \approx 0.0018658.$$

11. This is equivalent to asking at what effective expected return rate on the short-selling-allowed optimal portfolio(which includes T-Bill), do we achieve the same utility as the no-short-selling optimal portfolio(also includes T-Bill), fixing the variance of the short-selling-allowed optimal portfolio calculated from question 9. That is, we solve

$$r_{ss}^* - \frac{15(0.02214\%)}{2} = 0.0018326 \Rightarrow r_{ss}^* \approx 0.34931\%$$

The difference between r_{ss}^* and the actual expected return rate on the short-selling-allowed optimal portfolio is the maximum reduction in the expected return of the short-selling-allowed optimal portfolio such that the utility from it is still at least as high as the utility gained from the no-short-selling optimal portfolio:

$$\text{maximum reduction } = 0.35263\% - 0.34931\% = 0.00332\%$$

as wanted.